

Tasks Completion

Completed Tasks

2.1 Dataset Source

Dataset Sources were downloaded manually

2.3 Exploration Tasks

Explored the downloaded dataset

3.1 Data Ingestion & Profiling

Implemented a data download script, profiled the dataset, Assessed completion, Generated data quality report

3.2 Data Cleaning & Standardization

Standardized data columns, Handled missing values

3.4 Data Modeling

Designed the dimensional model, defined tables, Queried the DB for analysis

4.1 Summary Statistics & Distributions

Visualized price distribution across key factors, analyzed the listing count distribution, examined review scores, explored availability patterns

4.3 Temporal & Seasonal Trends

Examined review volumes, investigated minimum night policies

4.4 Host & Supply-Side Analysis

Segmented hosts by portfolio, compared pricing strategies, explored market supply concentration

4.5 Review & Demand-Side Analysis

Examined review count - price relationship,

Incomplete Tasks

3.3 Data Enrichment & Joining

Didn't complete due to prioritization. Same output was able to be produced via individual tables. Traded off listing enrichment for better results with available data

3.5 Pipeline Design & Automation

Didn't complete due to prioritization, Files were parsed and cleaned via scripts with manual steps. Traded off due to time constrictions

3.6 Advanced & Cloud-Native Topics

Didn't complete due to prioritization. Optional tasks which doesn't affect data analysis results

4.2 Geographic & Spatial Analysis

Didn't complete due to prioritization.

4.5 Review & Demand-Side Analysis

(Identify listings with high review counts but low scores and determine what characterizes them.)

Didn't complete due to lack of data in review scores.

5 Statistical Analysis

Didn't complete due to Time constrictions.

6 Data Science Challenges

Didn't complete due to Time constrictions.

7 AI and Machine Learning Opportunities

Didn't complete due to Time constrictions.

8 Open Innovation Challenge

Didn't complete due to Time constrictions.